FermiWick

Reference manual

Every function, option and convention, with an example computed by the package

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1 Installation and loading

FermiWick is a paclet. Install the paclet of the latest release once, then load it in every session, with the Gutzwiller and Fock modules when needed; the guided tour comes with the paclet:

```
PacletInstall["https://github.com/n4sd4q/FermiWick/releases/download/v0.6.0/FermiWick-0.6.0.paclet"]
Needs["FermiWick`"]; Needs["FermiWick`Gutzwiller`"]; Needs["FermiWick`Fock`"];
NotebookOpen[PacletObject["FermiWick"]["AssetLocation", "Examples"]]
```

To use a clone of the repository without installing, load its folder before `Needs`:

```
PacletDirectoryLoad["/path/to/FermiWick/FermiWick"]
```

FermiWick has no dependencies. It needs Mathematica 13.0 or later and is tested with 15.0.

Loaded in a notebook, the package prints a banner (the version, the license, how to cite it, links to the repository and the website) and looks for a newer release; a script prints nothing and never checks.

`FermiWickUpdateCheck, $FermiWickUpdateCheck`

At most once a day, in the background, loading the package in a notebook asks GitHub (`api.github.com`) for the latest release and, if it is newer than `$FermiWickVersion`, adds to the banner the `PacletInstall` line of that release; any failure is silent, and a session without access to the Internet never asks. `$FermiWickUpdateCheck = False` before `Needs["FermiWick`"]` (in `init.m` for good) turns the check off. `FermiWickUpdateCheck[]` asks by hand and gives the version loaded, the latest one and the line that installs it, or `$Failed` with a message.

The test suite runs with `math -script Tests/RunTests.wls`.

2 Conventions

- **Operators.** `c[l1, l2, ...]` and `cDag[l1, l2, ...]` annihilate and create a fermion in the mode with labels `l1, l2, ...`; `DeclareModeLabels` sets the meaning of each label. By default `c[i]` is a spinless fermion on site `i` and `c[i, s]` a fermion on site `i` with spin `s`. A site is any expression, and two different symbolic sites are different sites: `KroneckerDelta[i, j]` is 0. A spin is `SpinUp`, `SpinDown` or a symbol, split into ↑ and ↓ when needed. Generic labels (orbitals, momenta, ...) are compared with `KroneckerDelta`, which is kept and substituted in what it multiplies.
- **Products.** `a ** b` is the product in the given order; `a · b` (`\[CenterDot]`) is the product brought to normal order. Scalars (anything free of operators) factor out.
- **Normal order.** All $c^\dagger$ to the left of all c; results are expanded sums of monomials.
- **Contractions.** One amplitude for each: the hopping $\chi_{ij}^{\sigma\sigma'} = \langle c_{i\sigma}^\dagger c_{j\sigma'}\rangle$, `\[Chi][i, j, s, t]`, and the pairing $\Delta_{ij}^{\sigma\sigma'} = \langle c_{i\sigma} c_{j\sigma'}\rangle$, `\[CapitalDelta][i, j, s, t]`. A symmetry removes indices: with S^z conserved χ_{ij}^{σ} (`\[Chi][i, j, s]`) and $\Delta_{ij}^{\uparrow\downarrow}$, $\Delta_{ij}^{\downarrow\uparrow}$; with SU(2) χ_{ij} and the singlet $\Delta_{ij} = \langle c_{i\uparrow} c_{j\downarrow}\rangle$, symmetric in the sites. Hermiticity gives $\langle c_b^\dagger c_a\rangle = \langle c_a^\dagger c_b\rangle^*$ and antisymmetry $\langle c_b c_a\rangle = -\langle c_a c_b\rangle$: an amplitude is written with its modes in canonical order (the sites, then the spins, ↑ first), the other order as its conjugate or its opposite (`Conjugate[\[Chi][1, 2]]` for χ_{21}). On one site $\chi_{ii}^{\sigma\sigma}$ are the densities, $\chi_{ii}^{\uparrow\downarrow} = \langle S_i^+\rangle$ and $\Delta_{ii}^{\uparrow\downarrow} = \langle c_{i\uparrow} c_{i\downarrow}\rangle$; with SU(2), χ_{ii} is the density per spin. Modes with more labels group them by type, the spins last: `\[Chi][i, j, o, o', s, t]`.
- **Complex conjugation.** The amplitudes are complex; every other symbol is taken as real by `Dagger`, unless declared complex.

Symbolic spins and sites:

```
NormalOrder[c[i, σ] ** cDag[j, σ]]
```

$$-\delta_{\downarrow,\sigma} c_{j\downarrow}^\dagger c_{i\downarrow} - \delta_{\uparrow,\sigma} c_{j\uparrow}^\dagger c_{i\uparrow}$$

```
NormalOrder[c[i, σ] ** cDag[i, σ]]
```

$$\delta_{\downarrow,\sigma}(1 - c_{i\downarrow}^\dagger c_{i\downarrow}) + \delta_{\uparrow,\sigma}(1 - c_{i\uparrow}^\dagger c_{i\uparrow})$$

3 Operators

`c, cDag`

The ladder operators. Their square vanishes:

```
NormalOrder[cDag[1, "↑"] ** cDag[1, "↑"]]
```

```
0
```

`DeclareModeLabels, $ModeLabels`

`DeclareModeLabels[{type1, type2, ...}]` declares the labels of the operators with that number of labels: `"Site"` (locality: products are reduced site by site), `"Spin"` (a spin $\frac{1}{2}$; the spin operators below need modes `c[i, s]`), or any other name, a generic label. Defaults: `{"Site"}`, `{"Site", "Spin"}`, and `{"Site", "Label", ...}` for more labels. `$ModeLabels` holds the declarations. Modes with no site are normal-ordered without the site-by-site reduction.

```
DeclareModeLabels[{"Site", "Orbital", "Spin"}];
NormalOrder[c[1, x, SpinUp] ** cDag[1, y, SpinUp] ** c[2, x, SpinDown]]
```

$$c_{2,x,\downarrow}\delta_{x,y} - c^\dagger_{1,y,\uparrow}c_{1,x,\uparrow}c_{2,x,\downarrow}$$

`SpinUp, SpinDown, $Spins`

`SpinUp` and `SpinDown` are the spins $\uparrow$ and $\downarrow$, `$Spins` their list, for spin sums:

```
NormalOrder[Sum[cDag[1, s] ** c[2, s], {s, $Spins}] ** c[3, SpinUp]]
```

$$c^\dagger_{1\downarrow}c_{2\downarrow}c_{3\uparrow} + c^\dagger_{1\uparrow}c_{2\uparrow}c_{3\uparrow}$$

`nOp, nTotOp, SzOp, SpOp, SmOp, DOp, B, Bdag`

The local operators of a site with spin; they stay compact, and `ToLadderBasis` writes them with c, $c^\dagger$. All the composite operators and their definitions:

FermiWick	operator	definition
`nOp[i, s]`	$n_{i\sigma}$	$c^\dagger_{i\sigma}c_{i\sigma}$
`nTotOp[i]`, `nOp[i]`	n_i	$n_{i\uparrow} + n_{i\downarrow}$
`SzOp[i]`	S^z_i	$\frac{1}{2}\left(n_{i\uparrow} - n_{i\downarrow}\right)$
`SpOp[i]`, `SmOp[i]`	S^+_i, S^-_i	$c^\dagger_{i\uparrow}c_{i\downarrow}$, $c^\dagger_{i\downarrow}c_{i\uparrow}$
`SpinOperator[i, a]`	S^a_i	$\frac{1}{2}\sum_{\sigma\sigma'} c^\dagger_{i\sigma}\,\sigma^a_{\sigma\sigma'}\,c_{i\sigma'}$, $a = 1,2,3$
`DOp[i]`	D_i (double occupancy)	$n_{i\uparrow}n_{i\downarrow}$
`Bdag[i]`, `B[i]`	$B^\dagger_i$, B_i (doublon)	$c^\dagger_{i\uparrow}c^\dagger_{i\downarrow}$, $c_{i\downarrow}c_{i\uparrow}$
`EmptyProjector[i]`	P^0_i	$(1-n_{i\uparrow})(1-n_{i\downarrow}) = 1 - n_i + D_i$
`SingleOccupationProjector[i]`	P^1_i	$n_i - 2D_i$
`GutzwillerProjector[i]`	P^G_i	$\sqrt{2}\left[\cos\theta\, P^1_i + \sin\theta\,(P^0_i + D_i)\right]$
`SdotS[i, j]`	$\mathbf{S}_i \cdot \mathbf{S}_j$	$S^z_iS^z_j + \frac{1}{2}(S^+_iS^-_j + S^-_iS^+_j)$
`SdotCrossS[i, j, k]`	$\mathbf{S}_i \cdot (\mathbf{S}_j \times \mathbf{S}_k)$	$\sum_{abc}\epsilon_{abc}\,S^a_iS^b_jS^c_k$

Products of these on one site reduce with the algebra of the fermions (e.g. $D_in_i = 2D_i$, $(S^z_i)^2 = \frac{1}{4}n_i - \frac{1}{2}D_i$); operators on several sites and products with symbolic sites stay as products.

`GutzwillerProjector` is in `FermiWick`Gutzwiller`; one-site operators on several sites are products, `DOp[i, j]` $= D_i D_j$.

```
ToLadderBasis[SzOp[1]]
```

$$\frac{1}{2}c^\dagger_{1\uparrow}c_{1\uparrow} - \frac{1}{2}c^\dagger_{1\downarrow}c_{1\downarrow}$$

`SpinOperator, SdotS, SdotScrossS`

`SpinOperator[i, a]` is S^a_i, $a = 1,2,3$; `SdotS[i, j]` is $\mathbf{S}_i \cdot \mathbf{S}_j$ and `SdotScrossS[i, j, k]` the scalar chirality $\mathbf{S}_i \cdot (\mathbf{S}_j \times \mathbf{S}_k)$.

```
NormalOrder[Sum[SpinOperator[1, a] · SpinOperator[2, a], {a, 3}] - SdotS[1, 2]]
```

$$0$$

`EmptyProjector, SingleOccupationProjector`

$P^0_i = 1 - n_i + D_i$ and $P^1_i = n_i - 2D_i$, so that $P^0 + P^1 + D = 1$; with several sites they are products.

```
ToSpinChargeBasis[EmptyProjector[1] + SingleOccupationProjector[1] + DOp[1]]
```

$$1$$

`DefineOperator`

`DefineOperator[head, n, definition, display]` defines an operator on n sites: `definition` is a function of the sites written with known operators, `display` a function of the sites that gives its form on screen. The operator stays compact; `NormalOrder`, `WickVEV` and `MeanField` expand it, and `ToSpinChargeBasis` recognizes it again. Options: `"Recognize"` (default `True` for $n \geq 2$) and `"Required" -> All` (recognize it only when it is present whole; by default its diagonal part may leave a remainder, as in $J\,\mathbf{S}\cdot\mathbf{S} + (J_z - J)S^zS^z$). A one-site operator on several sites is the product. `$DefinedOperators` is the registry.

```
DefineOperator[Hop, 2, (cDag[#1, "↑"] ** c[#2, "↑"] + cDag[#1, "↓"] ** c[#2, "↓"]) &,
  Subscript["T", Row[{#1, #2}]] &];
ToSpinChargeBasis[NormalOrder[2 Hop[1, 2] ** nTotOp[3]]]
```

$$2\mathrm{T}_{12}n_3$$

`AdjointOperator`

The adjoint of a defined operator stays compact: `Dagger` gives the operator itself when it is Hermitian, the operator with its sites permuted when that is its adjoint (a hopping), and otherwise `AdjointOperator[head][sites]`, registered automatically, shown with a dagger, expanded and recognized like the operator itself. Here the hopping of the example above and an operator that annihilates a singlet pair on a bond:

```
DefineOperator[Pair, 2,
  (c[#1, "↑"] ** c[#2, "↓"] - c[#1, "↓"] ** c[#2, "↑"])/Sqrt[2] &,
  Subscript["b", Row[{#1, #2}]] &];
{Dagger[Hop[1, 2]], Dagger[Pair[1, 2]],
  ToSpinChargeBasis[Dagger[Pair[1, 2]] ** Pair[1, 2]]}
```

$$\left\{\mathrm{T}_{21},\ \mathrm{b}_{12}{}^{\dagger},\ \frac{1}{4}n_1 n_2 - \mathbf{S}_1\cdot\mathbf{S}_2\right\}$$

4 Products and bases

`NormalOrder`

Brings an expression to normal order. Products are separated site by site: each factor acts on one site and is even or odd, so the sites are reordered with the fermion sign only and each site is reduced once.

`CenterDot`

`a · b · ...` is the normal-ordered product.

```
ToSpinChargeBasis[SzOp[1] · SzOp[1]]
```

$$\frac{n_1}{4} - \frac{D_1}{2}$$

`OperatorCommutator, OperatorAnticommutator`

$ab - ba$ and $ab + ba$, in normal order.

```
OperatorAnticommutator[c[1, "↑"], cDag[1, "↑"]]
```

$$1$$

`ToSpinChargeBasis`

Writes an expression with n, S^z, $S^{\pm}$, D, B, $B^\dagger$ and recognizes the defined operators; `"Recognize" -> False` leaves them expanded.

```
ToSpinChargeBasis[cDag[1, "↑"] ** c[1, "↓"] ** cDag[2, "↓"] ** c[2, "↑"]]
```

$$S_1^+ S_2^-$$

```
ToSpinChargeBasis[SdotS[1, 2], "Recognize" -> False]
```

$$\frac{1}{2}S_1^- S_2^+ + \frac{1}{2}S_1^+ S_2^- + S_1^z S_2^z$$

`ToLadderBasis`

Writes the composite and the defined operators with c and $c^\dagger$.

`GroupLocalOperators`

Normal-orders and groups the ladder operators site by site, with the fermion sign.

```
GroupLocalOperators[cDag[1, "↑"] ** cDag[2, "↑"] ** c[2, "↑"] ** c[1, "↑"]]
```

$$c^\dagger_{1\uparrow} c_{1\uparrow} c^\dagger_{2\uparrow} c_{2\uparrow}$$

`OperatorCoefficients`

The association monomial → coefficient of an expression in normal order; the constant has the key 1. `OperatorCoefficients[expr, m]` is the coefficient of one monomial, written in any order. It reads the fields of a mean-field Hamiltonian.

```
OperatorCoefficients[2 x cDag[1, "↑"] ** c[2, "↑"] + y + c[2, "↑"] ** cDag[1, "↑"]]
```

```
<|1 -> y, cDag[1, "↑"]**c[2, "↑"] -> -1 + 2*x|>
```

```
OperatorCoefficients[2 x cDag[1, SpinUp] ** c[2, SpinUp], c[2, SpinUp] ** cDag[1,
    SpinUp]]
```

$$-2x$$

`Dagger`

The Hermitian conjugate: products reversed, $c \leftrightarrow c^\dagger$, $S^+ \leftrightarrow S^-$, $B \leftrightarrow B^\dagger$, coefficients conjugated. The amplitudes χ, Δ are complex, the other symbols real; option `"Complex" -> {...}` declares more complex symbols. Operators defined with `DefineOperator` stay compact (`AdjointOperator`).

```
Dagger[t E^(I \[Alpha]) cDag[1, SpinUp] ** c[2, SpinUp] + χ[1, 2] SpOp[1]]
```

$$S_1^- \chi_{12}^* + e^{-i\alpha} t c_{2\uparrow}^\dagger c_{1\uparrow}$$

`PlusHC, HermitianOperatorQ`

`PlusHC[expr]` is `expr` + `expr`†; `HermitianOperatorQ[expr]` checks `expr` = `expr`†. Options as `Dagger`.

```
PlusHC[g c[1, SpinUp] ** c[2, SpinDown], "Complex" -> {g}]
```

$$g c_{1\uparrow} c_{2\downarrow} + g^* c_{2\downarrow}^\dagger c_{1\uparrow}^\dagger$$

`CleanDeltas, ResetNormalOrder, $SpinSymbols`

`CleanDeltas` simplifies the `KroneckerDelta` of sites and spins and does the sums over the spin; `ResetNormalOrder[]` clears the caches; `$SpinSymbols` lists the symbols used as spins.

5 Gaussian states

A Gaussian state is an object that gives the contraction $\langle O_1 O_2 \rangle$ of two ladder operators. It is declared by its symmetries, or by rules for the contractions. `state["Symmetries"]`, `state["Description"]` and `state["Contraction"]` give its parts.

`GaussianState`

Gaussian states, in which every expectation value follows from the contractions of two operators, are also called quasi-free states (pure, as Slater or BCS states, or mixed, as Gibbs states of quadratic Hamiltonians). `GaussianState[]` is the general one: every contraction, one amplitude each, the hopping $\chi_{ij}^{\sigma\sigma'} = \langle c_{i\sigma}^\dagger c_{j\sigma'} \rangle$ and the pairing $\Delta_{ij}^{\sigma\sigma'} = \langle c_{i\sigma} c_{j\sigma'} \rangle$, and nothing fixed, not even the filling. `ContractionTable` shows them between the modes of some sites, here $1\uparrow$, $1\downarrow$, $2\uparrow$, $2\downarrow$. On one site the diagonal entries of $\langle c_a^\dagger c_b \rangle$ are the densities and the other one is $\langle S_1^+ \rangle = \chi_{11}^{\uparrow\downarrow}$; Hermiticity always holds: an amplitude is written with its modes in canonical order, the other order as its conjugate.

```
MatrixForm[ContractionTable[GaussianState[], {1, 2}, "Normal"]]
```

$$\begin{pmatrix} \chi_{11}^{\uparrow\uparrow} & \chi_{11}^{\uparrow\downarrow} & \chi_{12}^{\uparrow\uparrow} & \chi_{12}^{\uparrow\downarrow} \\ \chi_{11}^{\uparrow\downarrow *} & \chi_{11}^{\downarrow\downarrow} & \chi_{12}^{\downarrow\uparrow} & \chi_{12}^{\downarrow\downarrow} \\ \chi_{12}^{\uparrow\uparrow *} & \chi_{12}^{\downarrow\uparrow *} & \chi_{22}^{\uparrow\uparrow} & \chi_{22}^{\uparrow\downarrow} \\ \chi_{12}^{\uparrow\downarrow *} & \chi_{12}^{\downarrow\downarrow *} & \chi_{22}^{\uparrow\downarrow *} & \chi_{22}^{\downarrow\downarrow} \end{pmatrix}$$

The pairing $\langle c_a c_b \rangle$ is antisymmetric, so that on one site only $\langle c_{1\uparrow} c_{1\downarrow} \rangle$ is left:

```
MatrixForm[ContractionTable[GaussianState[], {1, 2}, "Anomalous"]]
```

$$\begin{pmatrix} 0 & \Delta_{11}^{\uparrow\downarrow} & \Delta_{12}^{\uparrow\uparrow} & \Delta_{12}^{\uparrow\downarrow} \\ -\Delta_{11}^{\uparrow\downarrow} & 0 & \Delta_{12}^{\downarrow\uparrow} & \Delta_{12}^{\downarrow\downarrow} \\ -\Delta_{12}^{\uparrow\uparrow} & -\Delta_{12}^{\downarrow\uparrow} & 0 & \Delta_{22}^{\uparrow\downarrow} \\ -\Delta_{12}^{\uparrow\downarrow} & -\Delta_{12}^{\downarrow\downarrow} & -\Delta_{22}^{\uparrow\downarrow} & 0 \end{pmatrix}$$

`GaussianState[symmetries, options]` is the state with the given symmetries. Each one removes contractions, or the indices that told them apart. With the particle number conserved, `"Charge"`, there is no pairing:

```
MatrixForm[ContractionTable[GaussianState["Charge"], {1, 2}, "Anomalous"]]
```

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

With S^z conserved, `"SpinRotationZ"` (rotations about z), the hopping keeps the spin, one amplitude for each, $\chi_{ij}^{\uparrow}$ and $\chi_{ij}^{\downarrow}$, and the pairing joins opposite spins, $\Delta_{ij}^{\uparrow\downarrow}$ and $\Delta_{ij}^{\downarrow\uparrow}$:

```
MatrixForm[ContractionTable[GaussianState["SpinRotationZ"], {1, 2}, "Normal"]]
```

$$\begin{pmatrix} \chi_{11}^{\uparrow} & 0 & \chi_{12}^{\uparrow} & 0 \\ 0 & \chi_{11}^{\downarrow} & 0 & \chi_{12}^{\downarrow} \\ {\chi_{12}^{\uparrow}}^{*} & 0 & \chi_{22}^{\uparrow} & 0 \\ 0 & {\chi_{12}^{\downarrow}}^{*} & 0 & \chi_{22}^{\downarrow} \end{pmatrix}$$

```
MatrixForm[ContractionTable[GaussianState["SpinRotationZ"], {1, 2}, "Anomalous"]]
```

$$\begin{pmatrix} 0 & \Delta_{11}^{\uparrow\downarrow} & 0 & \Delta_{12}^{\uparrow\downarrow} \\ -\Delta_{11}^{\uparrow\downarrow} & 0 & \Delta_{12}^{\downarrow\uparrow} & 0 \\ 0 & -\Delta_{12}^{\downarrow\uparrow} & 0 & \Delta_{22}^{\uparrow\downarrow} \\ -\Delta_{12}^{\uparrow\downarrow} & 0 & -\Delta_{22}^{\uparrow\downarrow} & 0 \end{pmatrix}$$

With spin rotations, `"SpinRotation"` (SU(2)), the hopping is the same for both spins, χ_{ij}, and the pairing is a singlet, $\Delta_{ij} = \langle c_{i\uparrow} c_{j\downarrow} \rangle = -\langle c_{i\downarrow} c_{j\uparrow} \rangle$, symmetric in the sites:

```
MatrixForm[ContractionTable[GaussianState["SpinRotation"], {1, 2}, "Normal"]]
```

$$\begin{pmatrix} \chi_{11} & 0 & \chi_{12} & 0 \\ 0 & \chi_{11} & 0 & \chi_{12} \\ \chi_{12}^{*} & 0 & \chi_{22} & 0 \\ 0 & \chi_{12}^{*} & 0 & \chi_{22} \end{pmatrix}$$

```
MatrixForm[ContractionTable[GaussianState["SpinRotation"], {1, 2}, "Anomalous"]]
```

$$\begin{pmatrix} 0 & \Delta_{11} & 0 & \Delta_{12} \\ -\Delta_{11} & 0 & -\Delta_{12} & 0 \\ 0 & \Delta_{12} & 0 & \Delta_{22} \\ -\Delta_{12} & 0 & -\Delta_{22} & 0 \end{pmatrix}$$

With time reversal, `"TimeReversal"` ($T = i\sigma^y K$), the components of opposite spins are conjugates of each other ($\chi_{ij}^{\downarrow} = \chi_{ij}^{\uparrow *}$, $\chi_{ij}^{\downarrow\uparrow} = -\chi_{ij}^{\uparrow\downarrow *}$, $\Delta_{ij}^{\downarrow\uparrow} = -\Delta_{ij}^{\uparrow\downarrow *}$, $\Delta_{ij}^{\downarrow\downarrow} = \Delta_{ij}^{\uparrow\uparrow *}$), on one site there is no magnetization, and with SU(2) the amplitudes are real:

```
MatrixForm[ContractionTable[GaussianState[{"SpinRotation", "TimeReversal"}], {1, 2},
    "Normal"]]
```

$$\begin{pmatrix} \chi_{11} & 0 & \chi_{12} & 0 \\ 0 & \chi_{11} & 0 & \chi_{12} \\ \chi_{12} & 0 & \chi_{22} & 0 \\ 0 & \chi_{12} & 0 & \chi_{22} \end{pmatrix}$$

The contractions on one site are the densities, the magnetization and the on-site pairing; the options fix them:

option	default	
`"Hopping"`, `"Pairing"`	χ, Δ	names of the amplitudes
`"Filling"`	`Automatic`	$\langle n_{i\sigma}\rangle = \nu$; `Automatic`: the amplitudes χ_{ii}^{σ} (χ_{ii} with SU(2))
`"Magnetization"`	`Automatic`	$\langle n_{i\uparrow/\downarrow}\rangle = \nu \pm m$; a vector `{mx, my, mz}` in the general state, or a function of the site; `Automatic`: none when the filling is fixed
`"OnSitePairing"`	`Automatic`	$\langle c_{i\uparrow} c_{i\downarrow}\rangle$
`"Hermitian"`	`True`	$\langle c_b^\dagger c_a\rangle = \langle c_a^\dagger c_b\rangle^*$; `False`: an amplitude of its own
`"Rules"`	`{}`	replacements applied to the results

A state with S^z conserved, at half filling with a magnetization m along z:

```
magnetic = GaussianState[{"SpinRotationZ", "Charge"}, "Filling" -> 1/2,
    "Magnetization" -> m];
MatrixForm[ContractionTable[magnetic, {1}, "Normal"]]
```

$$\begin{pmatrix} m + \frac{1}{2} & 0 \\ 0 & \frac{1}{2} - m \end{pmatrix}$$

`GaussianState[{{op1, op2} :> value, ...}]` declares the contractions by rules instead, with patterns in the operators. Pairs with no rule contract to zero, and $\langle c\, c^\dagger\rangle$ follows from $\langle c^\dagger c\rangle$ by the anticommutator when it has no rule of its own. Two sites with hopping g:

```
dimer = GaussianState[{{cDag[i_, s_], c[j_, t_]} :> KroneckerDelta[s, t] If[i === j,
    1/2, g]}];
WickVEV[nOp[1, "↑"] ** nOp[2, "↑"], dimer]
```

$$\frac{1}{4} - g^2$$

ContractionTable

`ContractionTable[state, {i, j, ...}]` shows all the contractions between the modes of the sites, $\langle c_a^\dagger c_b \rangle$ and $\langle c_a c_b \rangle$; with a third argument `"Normal"` or `"Anomalous"` it gives the matrix, with the modes in the order of `Tuples[{sites, $Spins}]`. The state invariant under spin rotations, the charge conserved, at filling ν:

```
MatrixForm[ContractionTable[GaussianState[{"SpinRotation", "Charge"}, "Filling" -> ν
    ], {1, 2}, "Normal"]]
```

$$\begin{pmatrix} \nu & 0 & \chi_{12} & 0 \\ 0 & \nu & 0 & \chi_{12} \\ \chi_{12}^* & 0 & \nu & 0 \\ 0 & \chi_{12}^* & 0 & \nu \end{pmatrix}$$

Contraction

`Contraction[state, op1, op2]`; `Contraction[op1, op2]` uses `$State`.

```
{Contraction[dimer, cDag[1, "↑"], c[2, "↑"]], Contraction[dimer, c[1, "↑"], cDag[2,
    "↑"]]}
```

$$\{g, -g\}$$

SpinSingletState

The state of most calculations, written short: `SpinSingletState[]` is `GaussianState[{"SpinRotation", "Charge"}, "Filling" -> 1/2]`, and `"Pairing" -> \[CapitalDelta]` drops the charge symmetry, with the on-site pairing `"OnSitePairing"` (default 0). Options `"Hopping"`, `"Filling"`, `"Pairing"`, `"OnSitePairing"`, `"Hermitian"`.

```
WickVEV[DOp[1], SpinSingletState["Filling" -> ν, "Pairing" -> Δ, "OnSitePairing" -> η
    ]]
```

$$\eta\eta^* + \nu^2$$

`StateSelect, LocalPairQ, AnomalousPairQ`

`StateSelect[state, crit]` keeps the contractions of the pairs with `crit[op1, op2]` true. It is a filter, not a symmetry: it gives a physical state when the criterion is a symmetry or a decoupling of the sites (`LocalPairQ`: uncorrelated sites), and otherwise a way to isolate a class of terms. `AnomalousPairQ` is true for two c or two $c^\dagger$.

```
WickVEV[nTotOp[1] ** nTotOp[2], StateSelect[SpinSingletState[], LocalPairQ]]
```

```
1
```

`$State`

The state used when none is given, `SpinSingletState[]` by default.

6 Wick's theorem and mean field

`WickVEV`

`WickVEV[expr, state]` is the expectation value by Wick's theorem: the expression is normal-ordered, and each monomial is the sum over its pairings with the sign of the permutation. Options: `"MaxOrder" -> m`, at most m contractions between different sites; `"NormalOrder" -> False`, Wick on the expression as it is.

```
WickVEV[SdotS[1, 2] ** SdotS[2, 3], "MaxOrder" -> 2]
```

$$-\frac{3}{16}\chi_{13}\chi_{13}^*$$

`VerifyNormalOrder`

0 when `WickVEV` with and without normal ordering agree.

```
VerifyNormalOrder[c[1, "↑"] ** cDag[2, "↑"] ** c[2, "↑"] ** cDag[1, "↑"]]
```

```
0
```

`MeanField`

`MeanField[expr, state]` is the Hartree–Fock–Bogoliubov Hamiltonian of `expr`: every normal-ordered product $O_1 \cdots O_{2n}$ becomes

$$\sum_{p<q} (-1)^{p+q+1} O_p O_q \left\langle \prod_{r \neq p,q} O_r \right\rangle - (n-1) \langle O_1 \cdots O_{2n} \rangle,$$

two operators kept out in every way and the rest replaced by its expectation value. The result is quadratic, has the same expectation value as `expr` and the same first variation with respect to the state: it is the Hamiltonian whose Gibbs state makes $\langle H \rangle - TS$ stationary among Gaussian states. Quadratic terms are unchanged; odd terms give zero. The density–density interaction (Hartree and Fock):

```
MeanField[nTotOp[1] ** nTotOp[2]]
```

$$-\chi_{12}^* c_{1\downarrow}^\dagger c_{2\downarrow} - \chi_{12}^* c_{1\uparrow}^\dagger c_{2\uparrow} - \chi_{12} c_{2\downarrow}^\dagger c_{1\downarrow} - \chi_{12} c_{2\uparrow}^\dagger c_{1\uparrow} + c_{1\downarrow}^\dagger c_{1\downarrow} + c_{2\downarrow}^\dagger c_{2\downarrow} + c_{1\uparrow}^\dagger c_{1\uparrow} + c_{2\uparrow}^\dagger c_{2\uparrow} + 2\chi_{12}\chi_{12}^* - 1$$

```
MeanField[cDag[1, "↑"] ** c[2, "↑"]]
```

$$c_{1\uparrow}^\dagger c_{2\uparrow}$$

7 LaTeX

`ToTeX`

The LaTeX of an expression: ladder and local operators, spin operators, amplitudes and spins; the operators defined with `DefineOperator` are written with their display.

```
ToTeX[SdotS[1, 2] - 2 DOp[1] + χ[1, 2] cDag[1, SpinUp] ** c[2, SpinUp]]
```

```
\chi_{12} c^\dagger_{1\uparrow}c_{2\uparrow}+\mathbf S_{1}\cdot\mathbf S_{2}-2
    D_{1}
```

`WickPairings`

`WickPairings[product, state]` lists the pairings of a product of ladder operators that do not vanish in the state, in Wick's theorem applied to the product as written: `"Sign"`, `"Pairs"` (positions in the product) and `"Value"` (the product of the contractions).

```
WickPairings[cDag[1, SpinUp] ** c[2, SpinUp] ** cDag[2, SpinUp] ** c[1, SpinUp]]
```

```
{<|"Sign" -> 1, "Pairs" -> {{1, 2}, {3, 4}}, "Value" -> Conjugate[χ[1, 2]]*χ[1,
    2]|>,
 <|"Sign" -> 1, "Pairs" -> {{1, 4}, {2, 3}}, "Value" -> 1/4|>}
```

`WickTeX`

`WickTeX[product, state]` writes Wick's theorem for a product in LaTeX: the expectation value, the sum of the pairings drawn as contractions, their values and the result. Options `"Values"` and `"Result"` (default `True`) leave out the last two. The contractions use `simpler-wick`: `\usepackage{simpler-wick}` in the preamble, and two LaTeX runs. Overlapping contractions are drawn at different heights.

```
WickTeX[cDag[1, SpinUp] ** c[2, SpinUp] ** cDag[2, SpinUp] ** c[1, SpinUp], "Values"
    -> False]
```

```
\langle c^\dagger_{1\uparrow} c_{2\uparrow} c^\dagger_{2\uparrow}
    c_{1\uparrow}\rangle = \wick{\c1 c^\dagger_{1\uparrow} \c1 c_{2\uparrow} \c1
    c^\dagger_{2\uparrow} \c1 c_{1\uparrow}} + \wick{\c2 c^\dagger_{1\uparrow} \c1
    c_{2\uparrow} \c1 c^\dagger_{2\uparrow} \c2 c_{1\uparrow}} = \chi_{12}
    \chi_{12}^*+\frac{1}{4}
```

$$\langle c^\dagger_{1\uparrow} c_{2\uparrow} c^\dagger_{2\uparrow} c_{1\uparrow}\rangle = \overbrace{c^\dagger_{1\uparrow} c_{2\uparrow}}\, \overbrace{c^\dagger_{2\uparrow} c_{1\uparrow}} + \overbrace{c^\dagger_{1\uparrow} c_{2\uparrow}\, c^\dagger_{2\uparrow} c_{1\uparrow}} = \chi_{12}\chi_{12}^* + \frac{1}{4}$$

8 Gutzwiller projectors

Loaded by `Needs["FermiWick`Gutzwiller`"]`.

`GutzwillerProjector, \[Theta]`

$P_i^G = \sqrt{2}\,[\cos\theta\, P_i^1 + \sin\theta\,(P_i^0 + D_i)]$, the half-filling Gutzwiller operator with the variational angle θ; $\theta = 0$ projects on single occupation.

```
ToSpinChargeBasis[GutzwillerProjector[1]] /. θ -> 0
```

$$\sqrt{2}n_1 - 2\sqrt{2}D_1$$

`LocalSandwich`

`LocalSandwich[P, {i, j, ...}, X]` is $P_i P_j \cdots X \cdots P_j P_i$, normal-ordered; `LocalSandwich[PL, sites, X, PR]` uses different operators on the two sides.

```
ToSpinChargeBasis[LocalSandwich[GutzwillerProjector, {1}, DOp[1]]]
```

$$2D_1 \sin^2(\theta)$$

`GutzwillerVEV, GutzwillerSimplify`

`GutzwillerVEV[expr, state]` is `WickVEV` to first order in the contractions between different sites (the hopping amplitudes of the state), simplified by `GutzwillerSimplify`.

```
GutzwillerVEV[LocalSandwich[GutzwillerProjector, {1}, nTotOp[1]]]
```

$$1$$

9 Lattices

`DefinePeierlsHopping, DefineFlux`

`DefinePeierlsHopping[t, \[CurlyPhi]]` makes $t_{ab} = t\,e^{i\varphi_{ab}}$ with $\varphi_{ba} = -\varphi_{ab}$; `DefineFlux[\[CurlyPhi], {i, j, k, ...}, \[CapitalPhi]]` fixes the sum of the phases around the oriented loop to Φ, eliminating one free phase.

```
DefinePeierlsHopping[t, φ];
DefineFlux[φ, {1, 2, 3}, Φ];
Simplify[Subscript[t, 1, 2] Subscript[t, 2, 3] Subscript[t, 3, 1]]
```

$$t^3 e^{i\Phi}$$

10 Matrices and exact diagonalisation

Loaded by `Needs["FermiWick`Fock`"]`, in a cell of its own (the names must be known before the code that uses them is read):

```
Needs["FermiWick`Fock`"]
```

Conventions. The modes of a space are ladder operators of FermiWick, numbered site by site, then orbital by orbital, the spin fastest: $\alpha = (\text{site}-1)\,n_o n_s + (\text{orbital}-1)\,n_s + \text{spin}$, the order of `Tuples[{sites, orbitals, spins}]` and of `ContractionTable`. A basis state is the string of the occupations $(n_1,\dots,n_M)$, with index $k = \sum_\alpha n_\alpha 2^{M-\alpha}$ (the first mode is the most significant bit): the order of `Tuples[{0, 1}, M]`. The Jordan–Wigner sign is $c_\alpha|n\rangle = (-1)^{\sum_{\beta<\alpha} n_\beta}|n-e_\alpha\rangle$. The matrices are `SparseArray` objects; inside a sector the basis states are in increasing order of k.

`FockSpace, FockSpaceQ`

`FockSpace[sites]` is the space of spin-$\frac{1}{2}$ fermions on the sites, with the modes `c[i, s]`; `FockSpace[sites, "Orbitals" -> {o1, ...}]` has the modes `c[i, o, s]`, and `"Spins" -> None` gives spinless modes `c[i]` (or `c[i, o]`); `"Spins" -> {s1, s2}` names the two spins, the first one up. `space["Modes"]`, `space["Count"]` and `space["Dimension"]` (2^M) are its properties.

```
FockSpace[{1, 2}, "Orbitals" -> {"s", "p"}]["Modes"]
```

```
{c[1, "s", "↑"], c[1, "s", "↓"], c[1, "p", "↑"], c[1, "p", "↓"], c[2, "s", "↑"],
    c[2, "s", "↓"],
 c[2, "p", "↑"], c[2, "p", "↓"]}
```

`ModeIndex`

`ModeIndex[space, c[l1, l2, ...]]` (or `cDag`) is the index of a mode; `ModeIndex[space, l1, l2, ...]` the same from the labels. A mode outside the space gives a message and `$Failed`.

```
{ModeIndex[FockSpace[{1, 2}], c[2, "↑"]], FockSpace[Range[3], "Spins" ->
    None]["Modes"]}
```

```
{3, {c[1], c[2], c[3]}}
```

FockSector, FockSectorQ

`FockSector[space, "N" -> n, "Sz" -> sz, "Counts" -> {{modes, n}, ...}]` keeps the basis states with n particles, the given S^z (two spins) and n particles in each set of modes (ladder operators or indices). Every operator is built directly in a sector, without the whole space: sectors with fixed N are enumerated bit by bit and indexed through the combinatorial number system, so that time and memory scale with their dimension; sectors without fixed N need the whole space and are allowed up to `$FockFullLimit` modes (24). `ClearFockCache[]` forgets the sectors built so far.

```
{FockSector[FockSpace[Range[6]], "N" -> 6, "Sz" -> 0]["Dimension"],
 FockSector[FockSpace[{1, 2}], "Counts" -> {{{c[1, "↑"], c[1, "↓"]},
    2}}]["Dimension"]}
```

```
{400, 4}
```

FockMatrix

`FockMatrix[expr, target]` is the matrix of an expression of FermiWick: ladder operators, local and composite operators, operators defined with `DefineOperator` (also `GutzwillerProjector`), products with ** in the order written (no normal order is needed), coefficients numbers or symbols. `target` is a space, a sector (the image sector is found from the change of N, S^z and of every count) or a pair `{fromSector, toSector}`. The labels of every operator must be those of modes of the space; a symbolic spin must be summed first (`Sum[..., {s, $Spins}]`).

```
Normal[FockMatrix[cDag[1, "↑"] ** c[2, "↑"], FockSector[FockSpace[{1, 2}], "N" -> 1]]]
```

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

```
Normal[FockMatrix[J SdotS[1, 2], FockSector[FockSpace[{1, 2}], "N" -> 2, "Sz" -> 0]]]
```

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & -\frac{J}{4} & \frac{J}{2} & 0 \\ 0 & \frac{J}{2} & -\frac{J}{4} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

FockExpectation

`FockExpectation[expr, v, sector]` is $\langle v|$ `expr` $|v\rangle$ for a normalized vector `v` of the sector (or space); an operator that leaves the sector gives 0. Here the singlet of two sites:

```
sec = FockSector[FockSpace[{1, 2}], "N" -> 2, "Sz" -> 0];
singlet = {0, 1, -1, 0}/Sqrt[2];
{FockExpectation[SdotS[1, 2], singlet, sec], FockExpectation[DOp[1], singlet, sec]}
```

$$\left\{-\frac{3}{4}, 0\right\}$$

StateToKet

`StateToKet[target, v]` writes a vector as kets of the local occupations (0, ↑, ↓, ↑↓ for spin $\frac{1}{2}$; the occupations of the modes otherwise), site by site.

```
StateToKet[sec, singlet]
```

```
Ket[{{"↓"}, {"↑"}}]/Sqrt[2] - Ket[{{"↑"}, {"↓"}}]/Sqrt[2]
```

OccupationBasis, BasisState

`OccupationBasis[target]` lists the occupation vectors of the basis states, in order; `BasisState[target, occupations]` is the basis vector with the given occupations.

```
{OccupationBasis[sec], Normal[BasisState[sec, {1, 0, 0, 1}]]}
```

```
{{{0, 0, 1, 1}, {0, 1, 1, 0}, {1, 0, 0, 1}, {1, 1, 0, 0}}, {0, 0, 1, 0}}
```

SpinSquared

`SpinSquared[target]` is $\mathbf{S}^2$ of the whole space or of a sector (two spins), built from S^+ and S^z: faster than the matrix of $\sum_{ij} \mathbf{S}_i \cdot \mathbf{S}_j$.

```
Eigenvalues[Normal[SpinSquared[sec]]]
```

```
{2, 0, 0, 0}
```

LowestStates

`LowestStates[H, k]` gives `{energies, vectors}` of the k lowest eigenstates of a Hermitian matrix: dense up to dimension `$FockDenseLimit` (3000), Arnoldi above, from several starting vectors when $k > 1$, so that degenerate multiplets are complete.

```
dimer = -Sum[PlusHC[cDag[1, s] ** c[2, s]], {s, $Spins}] + 2 Sum[(nTotOp[i] - 1)^2,
    {i, 2}];
N[Round[10^4 First[LowestStates[FockMatrix[dimer, sec], 2]]]/10^4]
```

```
{-0.8284, 0.}
```

LehmannGreen, GreenMatrix

`LehmannGreen[H, space, modes]` is the single-particle Green's function on the whole Fock space, in the Lehmann representation, $G_{ab}(z) = \sum_{nm}(w_n + w_m)\,\langle n|c_a|m\rangle\langle m|c_b^\dagger|n\rangle / (z - E_m + E_n)$, for the modes given (ladder operators or indices). H conserves N: an expression of the package, a matrix on the whole space, or a function sector → matrix. The result is an association with `"Poles"`, `"Residues"`, `"Modes"`, `"Temperature"` and, when they are defined, `"GroundEnergy"`, `"GroundSectors"`, `"Degeneracy"`, `"GrandPotential"`; `GreenMatrix[g, z]` evaluates it at a complex frequency. At zero temperature (the default) the weights average over the degenerate ground multiplet, $G_{ab}(z) = \langle c_a(z - H + E_0)^{-1}c_b^\dagger\rangle + \langle c_b^\dagger(z + H - E_0)^{-1}c_a\rangle$; the residues add up to the identity (the last entry below):

```
space = FockSpace[{1, 2}];
g = LehmannGreen[dimer, space, {c[1, "↑"]}];
{N[Round[10^4 g["GroundEnergy"]]/10^4], g["Degeneracy"], Max[Abs[Total[g["Residues"]]
    - IdentityMatrix[1]]] < 10^-12}
```

```
{-0.8284, 1, True}
```

```
N[Round[10^4 GreenMatrix[g, 0.5 I]]/10^4]
```

```
{{0. - 0.1237*I}}
```

`"Temperature" -> T`, a number or a symbol, weighs every state with $e^{-(E_n-\mu N_n)/T}/Z$, and `"ChemicalPotential" -> \[Mu]` uses $H - \mu N$. At a temperature the transitions between excited states add poles; the grand potential, the number of poles and the sum rule of the residues:

```
gT = LehmannGreen[dimer, space, {c[1, "↑"]}, "Temperature" -> 1/2];
{N[Round[10^4 gT["GrandPotential"]]/10^4], Length[gT["Poles"]],
    Max[Abs[Total[gT["Residues"]] - IdentityMatrix[1]]] < 10^-12}
```

```
{-1.0871, 8, True}
```

With symbols in H every sector is diagonalised exactly (small clusters) and the result is in closed form. At zero temperature the ground level must be the lowest for every value of the symbols that `"Assumptions"` allows. The Hubbard atom at half filling, whose Green's function does not depend on the temperature:

```
atom = U/2 (nTotOp[1] - 1)^2;
FullSimplify[GreenMatrix[LehmannGreen[atom, FockSpace[{1}], {c[1, "↑"]},
    "Temperature" -> T], z][[1, 1]]]
```

$$-\frac{4z}{U^2 - 4z^2}$$

Other options: `"N"` (the particle numbers that carry weight, `All`), `"UseSz"` (sectors of S^z when H conserves it, `Automatic`), `"Tolerance"` (10^{-9}: energies closer than this are one level), `"ResidueCutoff"` (10^{-12}).

PartitionFunction, GrandPotential

`PartitionFunction[H, space, T, \[Mu]]` is the grand-canonical partition function $Z = \operatorname{Tr} e^{-(H-\mu N)/T}$ over the whole Fock space ($\mu = 0$ when it is left out), for H as in `LehmannGreen`. With symbols in H, T or μ it is exact, a sum of exponentials over the levels, to be differentiated: the grand potential $\Omega = -T \log Z$, the entropy $-\partial\Omega/\partial T$, the particle number $-\partial\Omega/\partial\mu$, the compressibility, the specific heat. Options `"N"` (one number: the canonical ensemble), `"UseSz"`, `"Assumptions"`, `"Tolerance"`. The Hubbard atom, and its particle number at $\mu = 0$:

```
Z = PartitionFunction[atom, FockSpace[{1}], T, μ]
```

$$2e^{\frac{\mu}{T}} + e^{\frac{2\mu-\frac{U}{2}}{T}} + e^{-\frac{U}{2T}}$$

```
FullSimplify[T D[Log[Z], μ] /. μ -> 0]
```

```
1
```

`GrandPotential[H, space, T, \[Mu]]` is $-T \log Z$; with numbers it is computed from the lowest level, without overflow at a low temperature, and at $T = 0$ it is the lowest value of $H - \mu N$:

```
N[Round[10^4 {GrandPotential[dimer, space, 1/2], GrandPotential[dimer, space, 1/100],
    GrandPotential[dimer, space, 0]}]/10^4]
```

```
{-1.0871, -0.8284, -0.8284}
```

ParticleHoleOperator, TimeReversal

`ParticleHoleOperator[space, first, last]` is the unitary P with $P c_\alpha P^{-1} = c_\alpha^\dagger$ for the modes `first..last` (all by default): σ^x on each of them times $e^{-i\pi\sum_\alpha(\alpha-1)n_\alpha}$. `TimeReversal[space]` gives `<|"U" -> U, "Apply" -> f, "Inverse" -> finv|>` for the antiunitary $T = UK$ of spin-$\frac{1}{2}$ fermions, with `f[v] = U.Conjugate[v]`.

```
With[{ph = ParticleHoleOperator[space]}, Max[Abs[ph . FockMatrix[c[1, "↑"], space] .
    Transpose[ph] - FockMatrix[cDag[1, "↑"], space]]]]
```

```
0
```

SectorBlock

`SectorBlock[H, sector]` is the block of a matrix on the whole space that acts in the sector.

```
SectorBlock[FockMatrix[DOp[1] + DOp[2], space], sec] == FockMatrix[DOp[1] + DOp[2],
    sec]
```

```
True
```